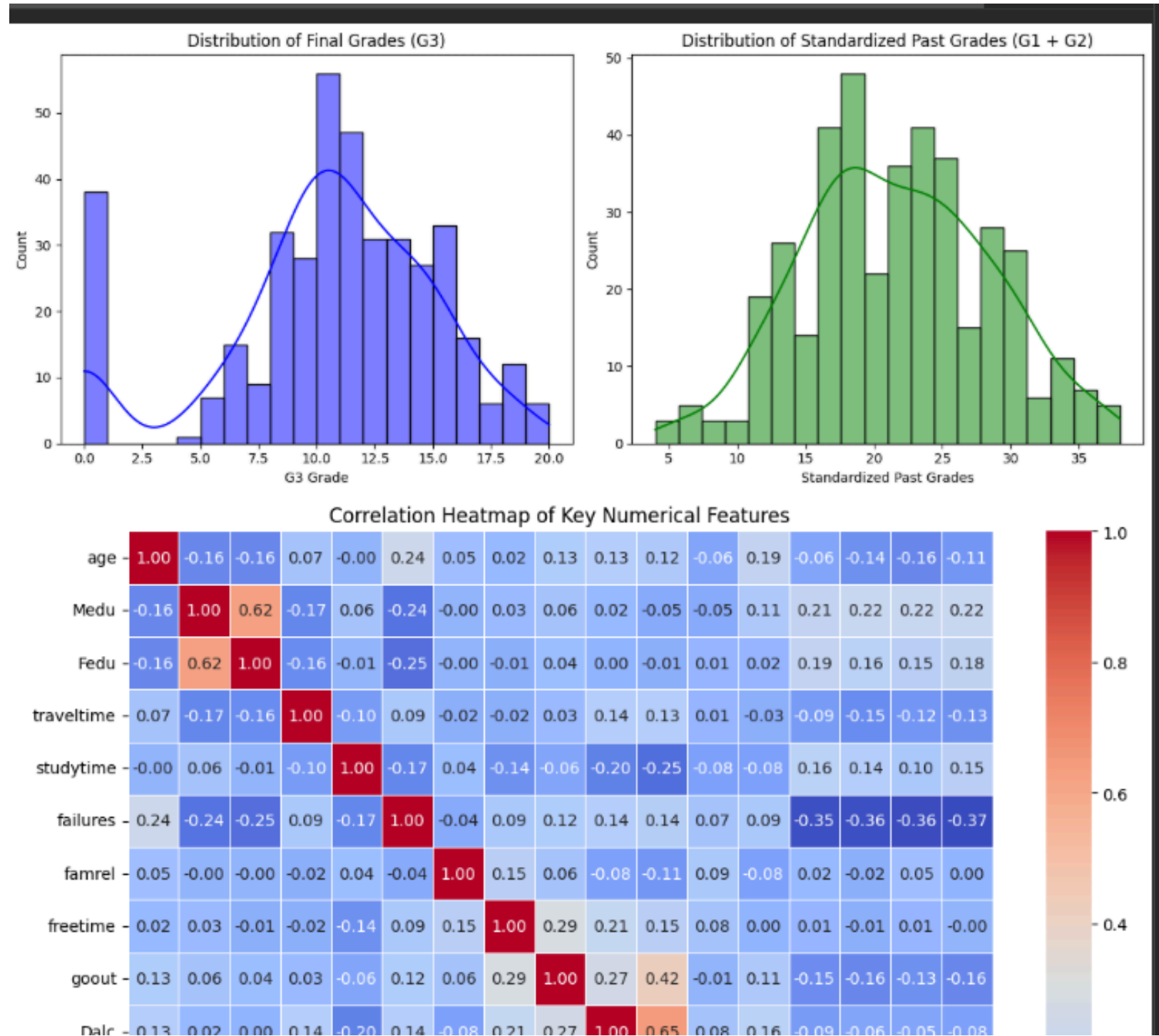


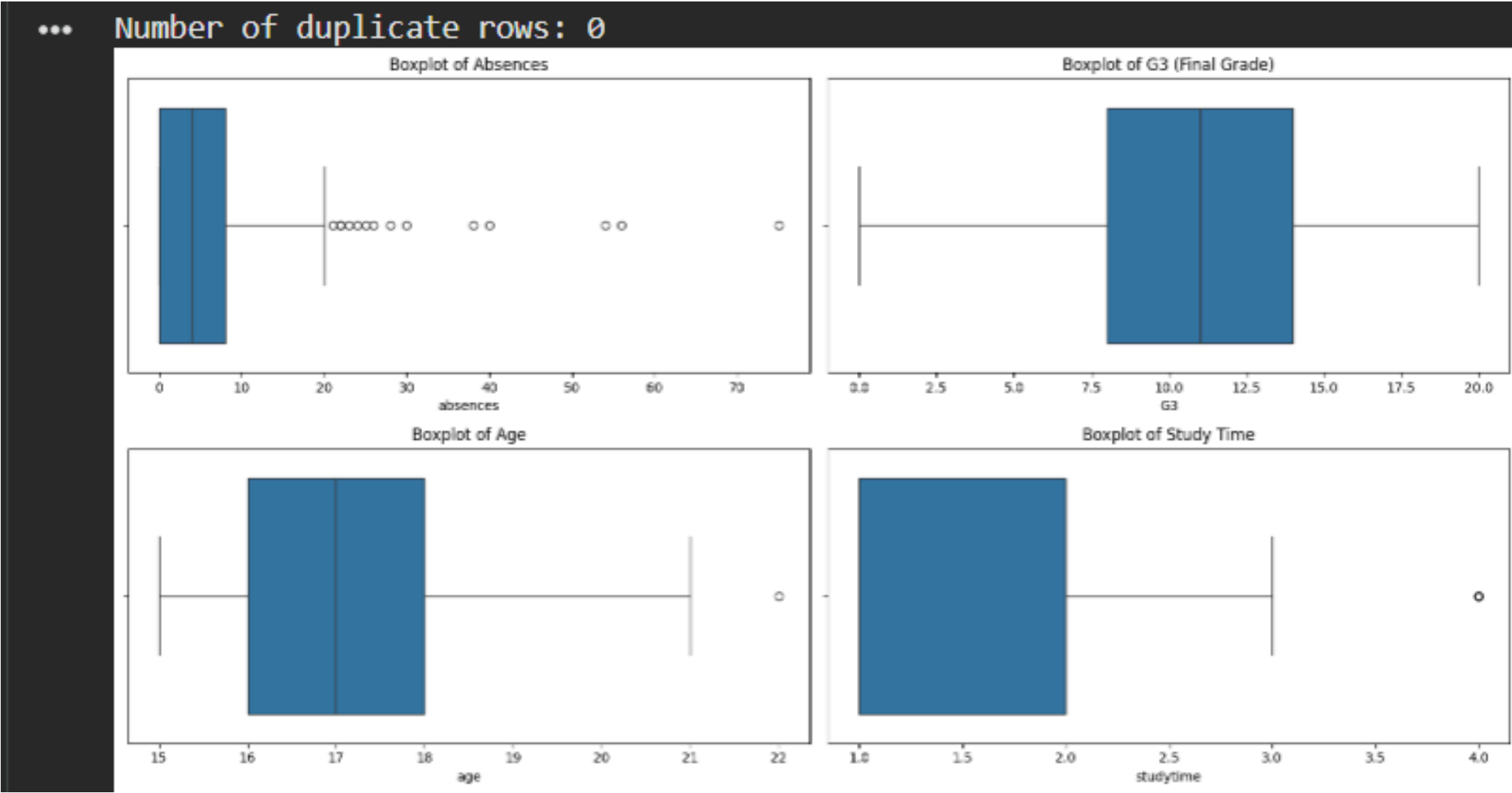
STUDENT  
PERFORMANCE  
ANALYSIS  
**DATA SCIENCE**

WIKTOR WŁODARCZYK  
ROSTYSLAV DMYTROVSKYI





1&2

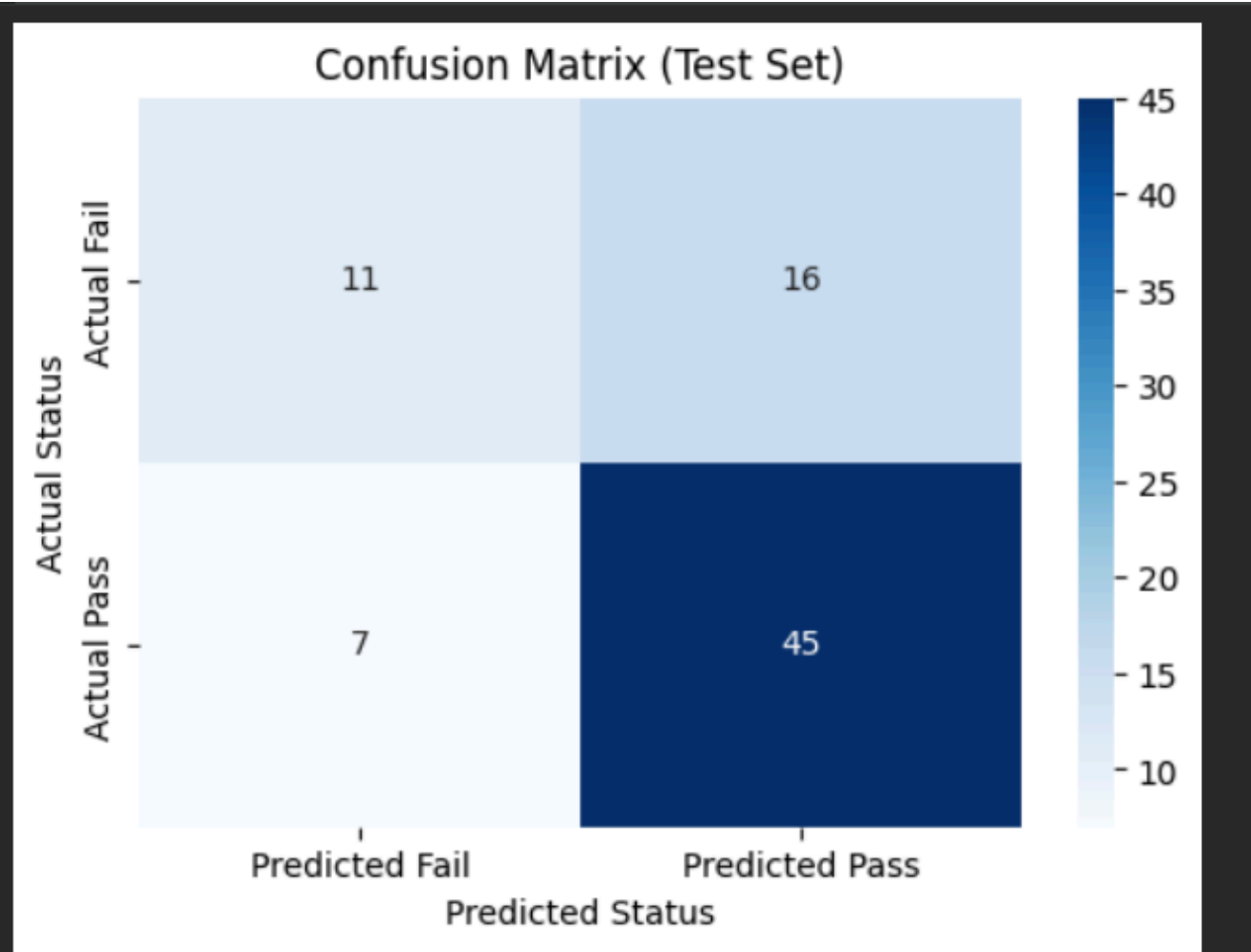


```
=== Model Performance Metrics (Test Set) ===
Mean Absolute Error (MAE): 1.6758 grade points
Mean Squared Error (MSE): 5.6647
Root Mean Squared Error (RMSE): 2.3801 grade points
Mean Absolute Percentage Error (MAPE): 413658.41%
Test Set R-squared (R2): 0.7237

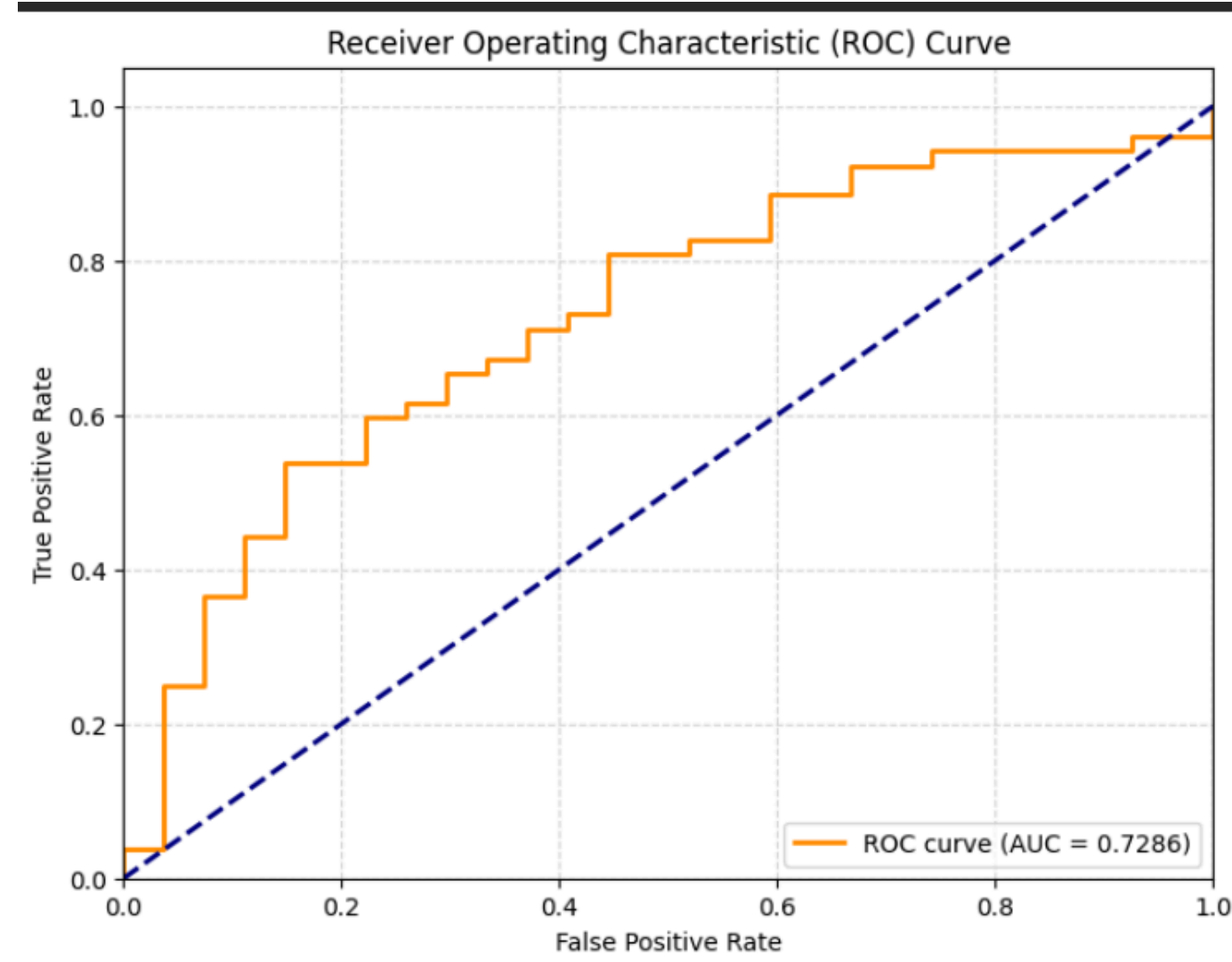
=== Sample Prediction Demonstration ===
Predicted G3 Grade: 6.12 / 20
Actual G3 Grade: 10.00 / 20
Absolute Error: 3.88 grade points
/tmp/ipykernel_2417/1299519760.py:12: SingularMatrixWarning: The de
ols model = sm.OLS(y_train, X_train const).fit()
```

# 3

## Regression



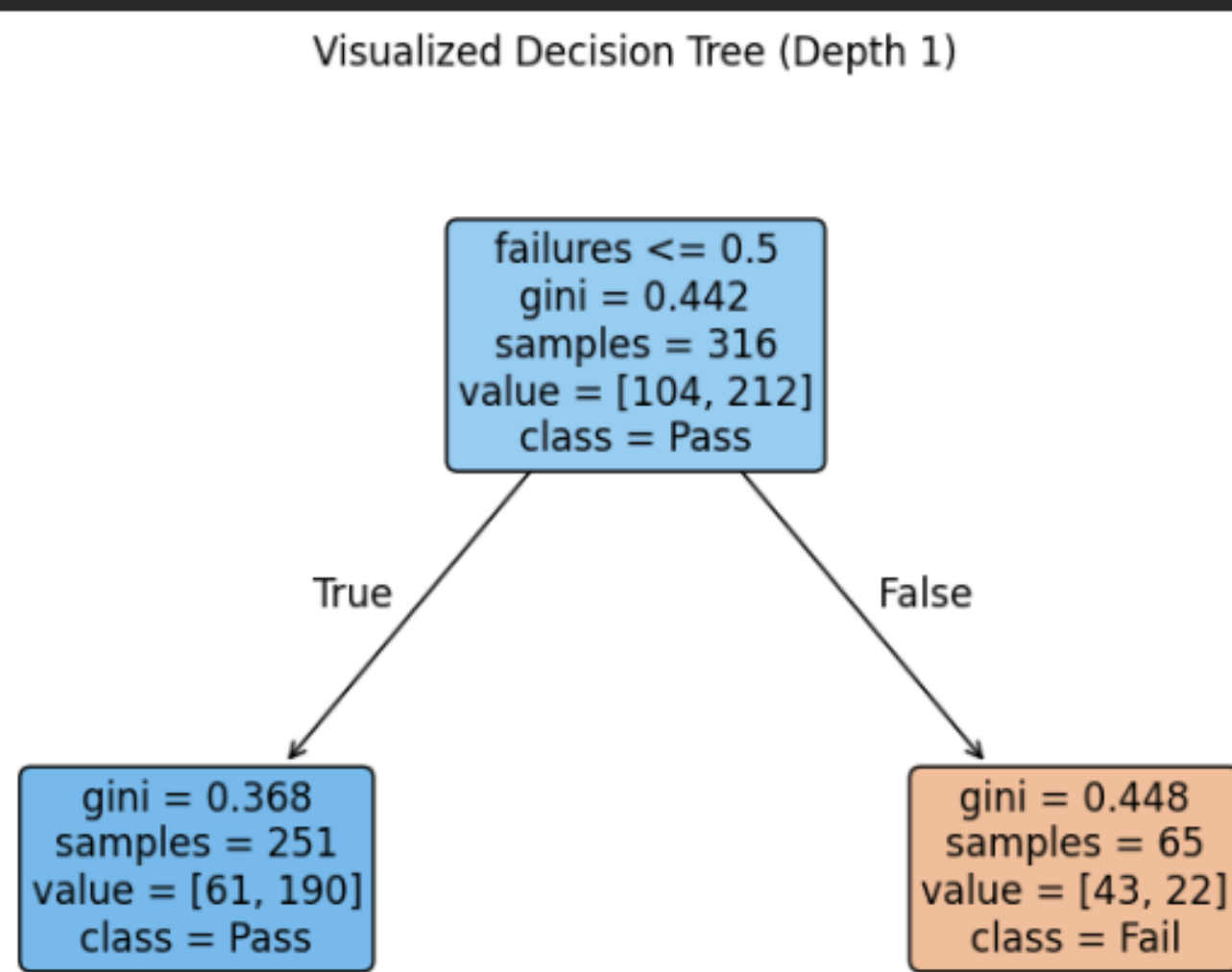
```
=== Classification Performance Metrics ===
Accuracy: 0.7089
Precision: 0.7377
Recall: 0.8654
F1-Score: 0.7965
ROC-AUC: 0.7286
```



```
=== Sample Probability Predictions ===
Actual_Class Predicted_Probability_Pass Predicted_Class
0 Pass 0.024742 Fail
```

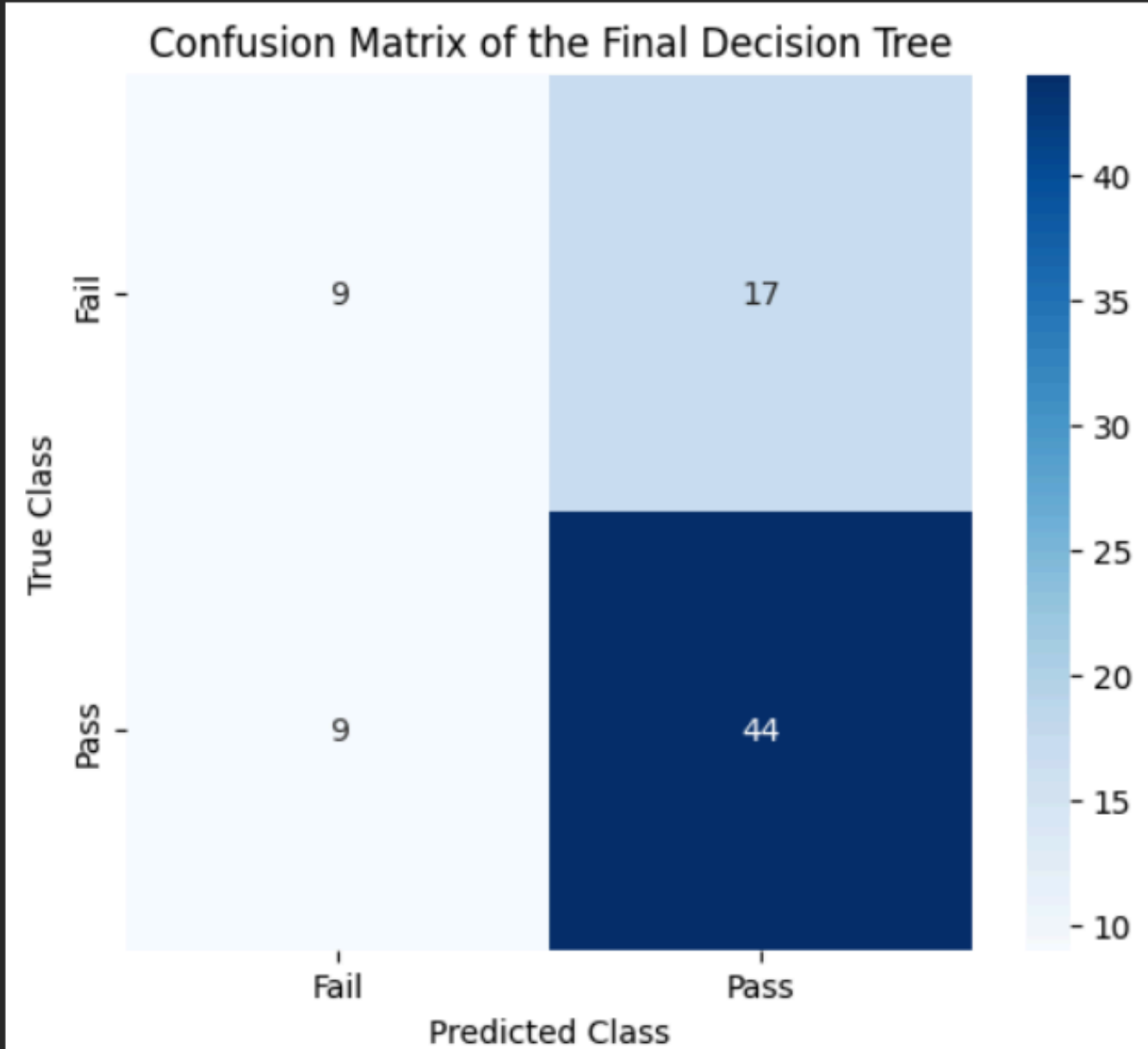
=== Final Model Classification Report ===

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Fail         | 0.50      | 0.35   | 0.41     | 26      |
| Pass         | 0.72      | 0.83   | 0.77     | 53      |
| accuracy     |           |        | 0.67     | 79      |
| macro avg    | 0.61      | 0.59   | 0.59     | 79      |
| weighted avg | 0.65      | 0.67   | 0.65     | 79      |



4

# decision trees



Top features used by the tree:

failures 0.636017

absences 0.175415

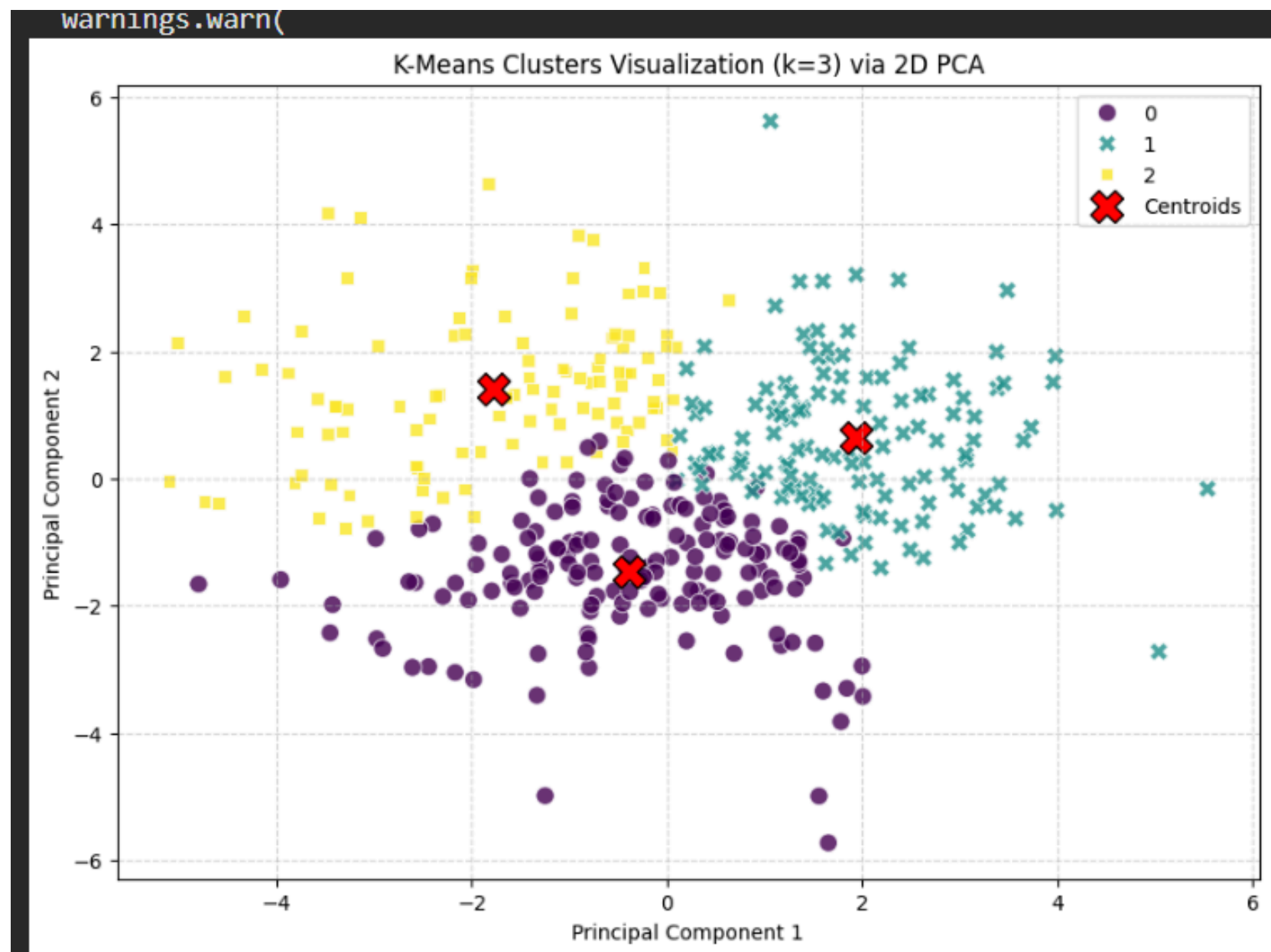
Mjob\_health 0.076993

dtype: float64

Decision Tree Accuracy: 0.6329

Decision Rule Classifier Accuracy: 0.6709

# k-means



=== Cluster Profiles (Feature Averages) ===

|         | G1        | G2        | G3        | studytime | freetime |
|---------|-----------|-----------|-----------|-----------|----------|
| cluster |           |           |           |           |          |
| 0       | 8.955975  | 8.522013  | 7.754717  | 2.169811  | 2.974843 |
| 1       | 14.269231 | 14.307692 | 14.469231 | 2.169231  | 3.176923 |
| 2       | 9.716981  | 9.594340  | 9.433962  | 1.669811  | 3.698113 |